DTU Compute
Department of Applied Mathematics and Computer Science

Occupancy-Aware Climate Control Estimating Thermal Comfort Curves with Hidden Markov Models

Bachelor Thesis



Occupancy-Aware Climate Control
Estimating Thermal Comfort Curves with Hidden Markov Models

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7 December, 2026

By
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Approval

This thesis has been prepared over six months at the Section for Indoor Climate, Department of Civil Engineering, at the Technical University of Denmark, DTU, in partial fulfilment for the degree Master of Science in Engineering, MSc Eng.

It is assumed that the reader has a basic knowledge in the areas of statistics.

Mads Holme Håkonsson - s234936

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Signature

.....

Date

Abstract

ABSTRACT IS MISSING

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Mads Holme Håkonsson, MSc Civil Engineering, DTU
Creator of this thesis template.

[Name], [Title], [affiliation]
[text]

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1 Introduction

Reduction in the usage of energy in buildings is a crucial aspect if one wants to go towards a more sustainable future. It has long been known that buildings are responseible for X% of the total energy consumption in the world, and that a large part of this energy is for HVAC systems [MISSING REFERENCE] HVAC systems energy consumption has been directly related to the occupant behaviours of the buildings. This imposes a challenge, as one has to be able to understand the behaviour of the occupants in the buildings before we can gain the largest optimization in energy consumption. (MISSING REFERENCE) has shown that there is a large variability in the behavour of occupants such as thermal comfort, lighting & ventiliation needs and precense hours.

Within the context of energy consumption in buildings, heating has been estimated to contribute to X% [MISSING REFERENCE] of the consumption. An aspect of heating of a building is the byproduct of the thermal comfort of the occupants in the building, and therefore, if we can understand the thermal comforts of the occupants in the buildings, we can optimize the energy consumption drasitically. However, the thermal comfort curves is a personal preference of the occupants, and we should therefore expect this to differer dependent on the individual buildings and the number of occupants.

This projects aims to understand when the occupants of a single offices are present, and how much thermal comfort feedback does we need to collect from the occupants in order to be able to infer the true thermal comfort of the occupants in the building. The goal is to also build a codebase others and easyly download and extent.
We will do this by using a Hidden Markov Model (HMM) to model the state of the room

WRITTEN OUTLINE OF EACH CHAPTER AND WHAT QUESTIONS THEY ANSWER

2 Data

The data used in this project are CO_2 data collected from the office building 303B and the censors and data providers are climify [MISSING REFERENCE]. The data is collected in the fall and winter period of 2023, and the raw data period spans from [MISSING REFERENCE] to [MISSING REFERENCE]. The data collected from the censors are in intervals of around 10 minutes. Because our HMM models works in discrete time, the data have been aggregated into 30 minutes such that each time step $\Delta t = 1$ represents 30 minutes.

When collecting CO_2 data, the censors tend to drift over time [MISSING REFERENCE], and therefore, the data collected from the censors are not always accurate. To empty rooms, that have the same CO_2 levels may show different readins based on how much the censors have drifted over time. This is a problem, as we want to use the CO_2 data to infer the presence of the occupants in the building and a higher baseline of the CO_2 data may lead to a false positive of the presence of the occupants in the building. To correct for this, we have used a method called *Automatic Baseline Correction* (ABC) Algorithm. [MISSING REFERENCE]. In the use of this algorithm, we have assumed that the lowest CO_2 ppm values over a period is equal to the outside CO_2 ppm values. From literature, it's common to set the outside CO_2 ppm values to 400 ppm [MISSING REFERENCE], which we have also done.

Let C_t be the CO_2 ppm value at time t, $\tilde{C}_t$ be the corrected CO_2 ppm value at time t.

We then define the modified ABC algorithm as

$$\tilde{C}_t = C_t - Q_{0.02}\left(C_{t-\Delta t:t}\right) + 400, \tag{2.1}$$

where $Q_{0.02}\left(C_{t-\Delta t:t}\right)$ is the 2% quantile of the CO_2 ppm values over the time period $[t - \Delta t, t]$ and Δt is a fixed time period. In this project, we have set $\Delta t = 8$ days, such that the weekday overlaps. The reason for using a 2% quantile is to avoid the effect of erradic readings of the CO_2 ppm values, which may occur if the censors are not working properly. Because we are interrested in predicting future presence of the occupants in the building,

we dont need the data to be as granular as 10 minutes.

Room	N	Min	2.5%	Median	Mean	97.5%	Max
Room 001	14289	341.00	426.00	621.00	828.52	2411.00	3185.00
Room 003	14328	335.00	405.00	516.00	611.92	1424.00	3025.00
Room 004	14268	292.00	400.00	940.50	1201.19	3042.00	3710.00
Room 005	14324	345.00	418.00	704.00	774.84	1500.00	2447.00
Room 007	14334	360.00	414.00	633.00	736.45	1588.00	2805.00
Room 008	14321	360.00	429.00	781.00	988.87	2824.00	3581.00
Room 009	14346	325.00	417.00	685.00	935.13	2322.50	3654.00
Room 012	14344	335.00	414.00	739.00	769.75	1406.00	2860.00
Room 013	14363	247.00	442.00	716.00	852.76	1841.00	2579.00
Room 014	14144	327.00	422.00	804.00	1098.36	3305.42	4000.00
Room 015	14336	337.00	414.00	754.50	985.40	2706.25	3808.00
Room 016	14333	349.00	422.30	686.00	833.00	1841.70	3825.00
Room 019	14338	355.00	435.00	727.00	886.71	1927.57	2756.00

Table 2.1: Summary statistics of the raw CO_2 readings (ppm) per room, after removing missing, saturated and duplicate readings.

TO DO:

- Make summary table of the data with before and after the ABC algorithm (n, mean, sd, min, quantiles, max, % of bins below 450 ppm; one row per room if several rooms).
- Make ACF plots to show the AR structure of the data. Add PACF (shows AR order); note daily cycle at lag 48.
- Make a plot of the data before and after the ABC algorithm (raw, estimated floor, calibrated + zoom on one week).
- Histogram/density of calibrated CO_2 (multimodality motivates hidden states).
- Average daily profile, weekday vs. off-day (motivates off-day + Fourier covariates).
- Overview plot of full series with train/val/test boundaries and feedback events.
- Short chapter introduction and conclusion.

Fix in existing text:

- Δt used both as 30-min step and as 8-day window: rename window to W.
- Dates: Room 009 segment is 2023-11-21 to 2024-02-09 (3860 bins, $\approx$ 80 days), not "fall and winter 2023". Clarify 303B vs. DTU dataset (live `room_009/30min` data comes from `clean_dtu_data.py`).
- Reason for 8 days: every window contains a full weekend (an empty-room period).
- Merge the duplicated aggregation paragraph; state that bins are means of the readings.
- Typos: censors $\rightarrow$ sensors, "To empty rooms" $\rightarrow$ Two, readins, erradic, interrested, dont, climify $\rightarrow$ Climify.
- Define a `\COtwo` macro instead of retyping `$\text{CO}_2$`.

Missing processing steps to describe:

- Raw cleaning: drop NaNs, keep 0–4000 ppm (sensor saturation), remove duplicates.
- Gap handling: only the longest gap-free segment is kept (discrete model needs equal spacing).
- ABC details: applied after 30-min binning (and why); trailing window, so no leakage into val/test; first 8 days dropped as warm-up; justify 2% quantile (must be below share of empty bins, $\approx$16% below 500 ppm in Room 009).
- Report floor drift (Room 009: 388 $\rightarrow$ 500 ppm).
- State ABC assumption: room is empty and ventilated to outdoor level at some point in every window. Outdoor CO_2 was $\approx$420 ppm in 2023–24: justify 400 or mention it.
- Feedback data: how collected, number of responses, binary flag per 30-min bin.
- Humidity: calibration against outdoor humidity, or state that it is excluded.

- Covariates: off-day flag (weekends + holidays), daily (3 harmonics) and weekly (1 harmonic) Fourier terms, weather variables; z-scored with training statistics only.
- Time zone: timestamps in UTC (no DST change in Nov–Feb).
- Split: chronological 65/15/20, boundaries 2024-01-12 and 2024-01-24; Christmas break falls in training.
- Rooms and sensors: which rooms and why, single/shared offices, ventilation type, sensor type (NDIR), accuracy, sampling interval.

Missing references: Climify, NDIR sensor drift, ABC (Senseair firmware; check whether your sensors already run it), outdoor CO_2 baseline (e.g. NOAA).

3 Theory

In this chapter, a ODE of the physical model of the CO_2 dynamics will be presented and how the process is descibed as a stochastic differential equation (SDE), where noise has been added to the system.
Given the presented SDE, it will be shown that the process can be desbribed as a discrete-time AR(1) process, where the production rate of the occupants in the room is modelled as a function of the occupancy state of the room.
Hereafter, the generel theory of the hidden markov models (HMM) will be presented along with the inference algorithms used in this project.

3.1 Modelling CO_2 dynamics

This subsection present the underlying physical model of the CO_2 dynamics of a room and show that it can be described as a discrete-time AR(1) process and how the production rate of the occupants in the room can be modelled as a function of the occupancy state of the room.

3.1.1 The underlying physical model & assumptions

Due to CO_2 does not decay by itself, the perfect mixtured mass balance model of a room (Batterman 2017; Wang, Burnett, and Chong 1999) is given as a first-order ordinary differential equation (ODE) as

$$V\frac{dC_{\text{indoor}}(t)}{dt} = G(t) + Q\left(C_{\text{outdoor}}(t) - C_{\text{indoor}}(t)\right), \tag{3.1}$$

where V is the room volume (m^3), C_{indoor} and C_{outdoor} are the indoor and outdoor CO_2 concentrations ($mg{\cdot}m^{-3}$), $G(t)$ is the occupant production rate ($mg{\cdot}h^{-1}$) and Q is the outdoor air flow rate ($m^3{\cdot}h^{-1}$).

Dividing by V gives

$$\frac{dC_{\text{indoor}}(t)}{dt} = \frac{G(t)}{V} - \frac{Q}{V}\left(C_{\text{indoor}}(t) - C_{\text{outdoor}}(t)\right). \tag{3.2}$$

Assuming C_{outdoor} is constant, the excess concentration $\tilde{C}(t) = C_{\text{indoor}}(t) - C_{\text{outdoor}}$ satisfies $d\tilde{C}/dt = dC_{\text{indoor}}/dt$, so

$$\frac{d\tilde{C}(t)}{dt} = \frac{G(t)}{V} - \frac{Q}{V}\tilde{C}(t). \tag{3.3}$$

The sensors report ppm, related by $C\,[\text{ppm}] = C\,[\text{mg}\cdot\text{m}^{-3}]\cdot V_m/M_{CO_2}$, with $M_{CO_2} = 44.01$ $g{\cdot}mol^{-1}$ and $V_m \approx 24.45$ $L{\cdot}mol^{-1}$ (25°C, 1 atm). Multiplying by the constant V_m/M_{CO_2} gives the model in ppm,

$$\frac{d\tilde{C}(t)}{dt} = P(t) - \alpha\tilde{C}(t), \qquad P(t) = \frac{G(t)\,V_m}{M_{CO_2}\,V}, \qquad \alpha = \frac{Q}{V}, \tag{3.4}$$

where $\tilde{C}(t)$ is now the excess CO_2 concentration (ppm), $P(t)$ is the production rate ($ppm{\cdot}h^{-1}$) and α is the air change rate (h^{-1}), i.e. the fraction of excess CO_2 ventilated out per hour.

The assumptions in 3.1, and therefore in 3.4, are that

- The CO_2 in the room is perfectly mixed, meaning that the CO_2 is evenly distributed in the room. This ensures that the sensors are not biased by location.
- There is no CO_2 sinks in the room, meaning that the CO_2 is not absorbed by any materials in the room and is only ventilated out of the room.

However, it's important to note that the model in 3.4 is a simplification of the real-world CO_2 dynamics of a room, and that there are many factors that can affect the CO_2 levels in a room, such as the number of occupants, the activity level of the occupants, the ventilation system, and the outdoor CO_2 levels.
One of the main limitations of the model is that it does not assume noise in the system, which is invenatiable as we use sensors.
Therefore, the next step to make the model more realistics is to add noise to the system in form of a stochastic differential equation (SDE), which will be presented in the next subsection.

3.1.2 The ODE as a diffusion Process (Ornstein-Uhlenbeck)

As previous described, the model in 3.4 desvibe the dynamics of the CO_2 in a room, but it does not take into account the noise in the system.
By adding noise to the system in form of a linear diffusion tern with constant diffusion coefficient σ, we can write the model as a stochastic differential equation (SDE) as

$$d\tilde{C(t)} = (P(t) - \alpha\tilde{C(t)})dt + \sigma dW(t), \tag{3.5}$$

where $W(t)$ is a standard Brownian motion.
3.5 is regornized as an Ornstein-Uhlenbeck process, which is a mean-reverting process, where the process tends to oscillate around some equilibrium state (chap. 1.13.1, Iacus 2008)

It is important to stress that the noise term in 3.5 was assumed to be a linear diffusion term with constant diffusion coefficient σ, as this is the simpelst noise to add to the system. However, one could also add a non-linear difussion term, where the diffusion coefficient is a function of the state of the system. This is however out of the scope of this project, as this would require more in depth knowledge of SDEs and stochastic calculus, which is not the focus of this project.

Next we want to derive some important properties of our SDE in 3.5, which will be used to show that the process can be described as a discrete-time AR(1) process.

3.1.3 Proof of AR(1) process

Given our model of the CO_2 dynamics with noise in 3.5, we now want to show that the process can be described as a discrete-time AR(1) process.
Firstly, we recognize the diffusion process as a Ornstein-Uhlenbeck, and the solution of this process is given by Iacus (2008, Sec. 1.13.1) as

$$\tilde{C}(t) = \frac{P}{\alpha} + \left(\tilde{C}_0 - \frac{P}{\alpha}\right) e^{-\alpha t} + \sigma \int_0^t e^{-\alpha(t-u)}\, \mathrm{d}W_u \tag{3.6a}$$

or, in the second parametrization,

$$\tilde{C}(t) = \mu + (\tilde{C}_0 - \mu)e^{-\alpha t} + \sigma \int_0^t e^{-\alpha(t-u)}\, \mathrm{d}W_u, \qquad \mu = \frac{P}{\alpha}, \tag{3.6b}$$

where $\tilde{C}_0$ is the initial value of the process at time $t = 0$. μ should be interpretated as the equilibrium state of the process, which is the state that the process tends to oscillate around.

If we then discretize the process with a time step Δt, we can write the process as

$$\tilde{C}_{t+\Delta t} = \mu + (\tilde{C}_0 - \mu)e^{-\alpha(t+\Delta t)} + \sigma \int_0^{t+\Delta t} e^{-\alpha(t+\Delta t-u)} \, \mathrm{d}W_u, \quad (3.7a)$$

where we can split the integral into two parts as

$$\sigma \int_0^{t+\Delta t} e^{-\alpha(t+\Delta t-u)} \, \mathrm{d}W_u = \sigma \Bigg(e^{-\alpha \Delta t} \int_0^t e^{-\alpha(t-u)} \, \mathrm{d}W_u + \int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)} \, \mathrm{d}W_u \Bigg). \quad (3.7b)$$

We can then write out the process as

$$\begin{aligned} \tilde{C}_{t+\Delta t} &= \mu(1 - e^{-\alpha \Delta t}) + \mu e^{-\alpha \Delta t} + (\tilde{C}_0 - \mu)e^{-\alpha t} e^{-\alpha \Delta t} \\ &+ \sigma e^{-\alpha \Delta t} \int_0^t e^{-\alpha(t-u)} \, \mathrm{d}W_u + \sigma \int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)} \, \mathrm{d}W_u \end{aligned} \quad (3.7c)$$

We see then that

$$\begin{aligned} \tilde{C}_{t+\Delta t} &= \left(\mu + (\tilde{C}_0 - \mu)e^{-\alpha t} + \sigma \int_0^t e^{-\alpha(t-u)} \, \mathrm{d}W_u \right) e^{-\alpha \Delta t} \\ &\quad + \mu(1 - e^{-\alpha \Delta t}) + \sigma \int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)} \, \mathrm{d}W_u \\ &= \tilde{C}_t e^{-\alpha \Delta t} + \mu(1 - e^{-\alpha \Delta t}) + \sigma \int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)} \, \mathrm{d}W_u \end{aligned} \quad (3.7d)$$

We have now shown that diffusion process is dependent on the previous emission $\tilde{C}_t$. But for it to be an AR(1) process, we nedd to show that the drift term has mean 0, constant variance and is unoccorelated over time. [MISSING REFERENCE].
Let $\Delta t = c \in \mathbb{N}$ be a constant time step and given that $\alpha, t \in \mathbb{R}_{\geq 0}$, then from 3.7d, we can see that the drift term is given by

$$\sigma \int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)} \, \mathrm{d}W_u = \quad (3.8)$$

We recognize $e^{-\alpha(t+\Delta t-u)}$ as our stochastic term, which is determinitics, meaning that that its ito intergratable as $\int_0^T e^{-2\alpha(t+\Delta t-u)} \, \mathrm{d}W_u < +\infty$.
Therefore, from Iacus (2008, Thm. 1.9.1), We now know that the mean of the drit term is

$$\mathbb{E}\left[\sigma \int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)} \, \mathrm{d}W_u \right] = 0 \quad (3.9)$$

And that the variance is

$$\mathrm{Var}\left[\sigma \int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)} \, \mathrm{d}W_u \right] = \sigma^2 \int_t^{t+\Delta t} e^{-2\alpha(t+\Delta t-u)} \, \mathrm{d}u \quad (3.10)$$

$$\mathrm{Var}\left[\sigma \int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)}\, \mathrm{d}W_u\right] = \sigma^2 \int_t^{t+\Delta t} e^{-2\alpha(t+\Delta t-u)}\, \mathrm{d}u \tag{3.11a}$$

$$= \sigma^2 e^{-2\alpha(t+\Delta t)} \int_t^{t+\Delta t} e^{2\alpha u}\, \mathrm{d}u \tag{3.11b}$$

$$= \sigma^2 e^{-2\alpha(t+\Delta t)} \left[\frac{1}{2\alpha} e^{2\alpha u}\right]_t^{t+\Delta t} \tag{3.11c}$$

$$= \frac{\sigma^2}{2\alpha} e^{-2\alpha(t+\Delta t)} \left(e^{2\alpha(t+\Delta t)} - e^{2\alpha t}\right) \tag{3.11d}$$

$$= \frac{\sigma^2}{2\alpha} \left(1 - e^{-2\alpha\Delta t}\right) \tag{3.11e}$$

We see that the variance is independent of t and that its constant over time.
We now only need to show that the covariance of the drift is 0.
We define that time steps as $t_1, t_2 \in \mathbb{R}_{\geq 0}$ such that $t_1 < t_2$ and that the time steps are disjoint, meaning that $[t_1, t_1 + \Delta t] \cap [t_2, t_2 + \Delta t] = \emptyset$ and that $t_1 + \Delta t = t_2$. We then define e_k as the drift term at time t_k such that

$$\begin{aligned} \mathbb{E}\left[e_k\right] &= 0 \\ \mathrm{Var}(e_k) &= \sigma \int_{t_{k-1}}^{t_k} e^{-\alpha(t+\Delta t-u)}\, \mathrm{d}W_u = \frac{\sigma^2}{2\alpha}\left(1 - e^{-2\alpha\Delta t}\right) \end{aligned} \tag{3.12}$$

We now then see that the covariance of the drift term is given by, given that $j \neq k$

$$\begin{aligned} \mathrm{Var}(e_k + e_j) &= \mathrm{Var}(e_k) + \mathrm{Var}(e_j) + 2\mathrm{Cov}(e_k, e_j) \\ &\Leftrightarrow \end{aligned} \tag{3.13}$$

To compute $\mathrm{Var}(e_k + e_j)$, we write both drift terms as integrals over the same interval $[0, T]$, where $T \geq \max(t_k, t_j) + \Delta t$, by introducing the deterministic functions

$$f_k(u) = \mathbb{1}_{[t_k,\, t_k+\Delta t)}(u)\, e^{-\alpha(t_k+\Delta t-u)}, \tag{3.14}$$

such that

$$e_k = \sigma \int_0^T f_k(u)\, \mathrm{d}W_u. \tag{3.15}$$

Since $j \neq k$, the intervals $[t_k, t_k + \Delta t)$ and $[t_j, t_j + \Delta t)$ are disjoint, and therefore

$$f_k(u) f_j(u) = 0 \quad \forall u \in [0, T]. \tag{3.16}$$

The sum $e_k + e_j = \sigma \int_0^T \left(f_k(u) + f_j(u)\right) \mathrm{d}W_u$ is again an Itô integral with a deterministic integrand, so by Iacus (2008, Thm. 1.9.1)

$$\begin{aligned} \mathrm{Var}(e_k + e_j) &= \sigma^2 \int_0^T \left(f_k(u) + f_j(u)\right)^2 \mathrm{d}u \\ &= \sigma^2 \int_0^T f_k(u)^2\, \mathrm{d}u + \sigma^2 \int_0^T f_j(u)^2\, \mathrm{d}u + 2\sigma^2 \int_0^T f_k(u) f_j(u)\, \mathrm{d}u \\ &= \mathrm{Var}(e_k) + \mathrm{Var}(e_j) + 0. \end{aligned} \tag{3.17}$$

Comparing with $\mathrm{Var}(e_k + e_j) = \mathrm{Var}(e_k) + \mathrm{Var}(e_j) + 2\,\mathrm{Cov}(e_k, e_j)$ gives

$$\mathrm{Cov}(e_k, e_j) = 0, \qquad j \neq k. \tag{3.18}$$

Hence the drift terms have mean zero, constant variance and are uncorrelated, and the discretized process is an AR(1) process.

3.1.4 Modelling Production Rate

By assuming that the prodction rate of the occupants is only depented on the amount of occupants in the room, we can model the production rate as a function of the occupancy state of the room. Let $S(t)$ be the occupancy state of the room at time t, then we can model the production rate as

$$P(t) = \beta_{S(t)}, \tag{3.19}$$

where $\beta_{S(t)}$ is the production rate of the occupants in the room, which is dependent on the occupancy state of the room. This gives a stepwise prodiuction rate function, which is only dependnt on the occupancy state of the room.

3.2 Hidden Markov Model (HMM) theory

This section will present the underlying models and the theory behind it that models the diffusion process. Here to, inference algorithms and analysis methods will be presented.

3.2.1 The underlying theory

Hidden markov models are markov chains where the underlying states are hidden and only a random variable that is dependent on the underlying state is observed.
This can be described as a markov chain with a set of hidden states $S = \{s_1, s_2, ..., s_N\}$ and a set of observations $O = \{o_1, o_2, ..., o_M\}$. The transition between the hidden states is governed by a transition matrix Γ and the observations are generated by an emission matrix B. The initial state distribution is given by π. The HMM stucture can then debsrib the emissions as

$$P(O_t|S_t) = B_{S_t,O_t}, \tag{3.20}$$

where B_{S_t,O_t} is the probability of observing O_t given the hidden state S_t. The transition between the hidden states is given by

$$P(S_t|S_{t-1}) = \Gamma_{S_{t-1},S_t}, \tag{3.21}$$

where Γ_{S_{t-1},S_t} is the probability of transitioning from hidden state S_{t-1} to hidden state S_t. The initial state distribution is given by

$$P(S_1) = \pi_{S_1}, \tag{3.22}$$

where π_{S_1} is the probability of starting in hidden state S_1.

3.2.2 Likelihood

Several methods can be used to estimate the parameters of a HMM, where at some of the more common methods are the Baum-Welch algorithm [MISSING REFERENCE].
In this project, we will however use the method of maximum likelihood estimation (MLE) to estimate the parameters of the HMM with the forward-filtering algorithm [MISSING REFERENCE].

The forward algorithm implemented is called recursive filtering (Møller et al. 2024), as it constructs the current state distribution $p(X_t \mid Y_{1:t})$ recursively from the previous state distribution $p(X_t \mid Y_{1:t-1})$.

Algorithm 1 Recursive Filtering Forward Algorithm

Require: Observations $\mathbf{y} = (y_1, \ldots, y_T)$, covariates $\mathbf{x} = (x_1, \ldots, x_T)$, initial state distribution $u_0 \in \mathbb{R}^{1 \times N}$, HMM parameters θ

Ensure: Predicted distributions $\{u_t\}$, filtered distributions $\{u_{t|t}\}$, step likelihoods $\{f_t\}$

for $t = 1, \ldots, T$ **do**
  $\Gamma_t \leftarrow \text{TransitionMatrix}(\theta, t, y_t, x_t)$ ▷ $N \times N$ transition matrix
  $u_{t|t-1} \leftarrow u_{t-1|t-1} \cdot \Gamma_t$ ▷ Predicted state distribution, $1 \times N$
  $g_t \leftarrow \text{Density}(\theta, t, y_t, x_t)$ ▷ Emission densities for all states, $1 \times N$
  $f_t \leftarrow \sum_{i=1}^{N} [u_t \odot g_t]_i$ ▷ Marginal likelihood at time t
  $f_t \leftarrow \max(f_t, 10^{-10})$ ▷ Clip to avoid division by zero
  $u_{t|t} \leftarrow \dfrac{u_t \odot g_t}{f_t}$ ▷ Filtered state distribution (normalized)
end for
return $\{u_t\}$, $\{u_{t|t}\}$, $\{f_t\}$

where $\odot$ denotes element-wise multiplication of vectors. The log-likelihood is obtained by summing the log of the step likelihoods:

$$\mathcal{L}(\theta) = \sum_{t=1}^{T} \log f_t. \tag{3.23}$$

Each likelihood step f_t is the sum of the predicted state distribution u_t multiplied by the emission densities g_t for all states, which gives the marginal likelihood of observing y_t given the previous observations $y_{1:t-1}$.

3.2.3 Forward Backward Filtering
3.2.4 Backward Sampling
3.2.5 Model Diagnostics

4 Implementation

5 Model Results

6 Simulation Study

A

7 Discussion

8 Conclusion

9 Future Work

Bibliography

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